NEF ABUNDANCE: FRAMEWORK – K-TRIVIAL, GENERAL DIMENSION

1. Theorem and proof

Fix an integer $d \geq 2$. We use the following source inputs, denoted collectively by (I_d).

- For every projective klt $\mathbb{Q}$-pair (Y, B) of dimension $j < d$ and every nef rational Cartier divisor M, if $K_Y + B$ is pseudo-effective and, for some $a \in \mathbb{Q}_{>0}$,

$$(K_Y + B + aM) \cdot C > 0$$

  for every integral curve through a sufficiently very general point, then $K_Y + B + cM$ is big for every $c \in \mathbb{Q}_{>0}$.
- Every projective log canonical $\mathbb{Q}$-pair of dimension at most d with nef adjoint has semiample adjoint.
- Every projective $\mathbb{Q}$-factorial generalized klt $\mathbb{Q}$-pair of dimension at most d, with fixed nef rational Cartier b-divisor data and pseudo-effective adjoint, admits a terminating choice of minimal-model program with a nef endpoint. The program extracts no divisors, preserves $\mathbb{Q}$-factorial generalized klt models, and on a common resolution the initial adjoint equals the final adjoint plus an effective divisor exceptional over the final model.

The first input is (P_j) of [1, Section 3.1]. The last two are the companion inputs used in [1, Section 2].

Theorem. *Assume* (I_d). *Let* X *be a projective* $\mathbb{Q}$*-factorial terminal variety over* $\mathbb{C}$ *of dimension* d*, with* $K_X \sim 0$*. Let* D *be a nef Cartier divisor with* $\mathrm{n}(X, D) = d$*. Then* D *is big. In particular,*

$$H^0(X, \mathcal{O}_X(mD)) \neq 0 \qquad \text{for some integer } m > 0.$$

Proof. Suppose for contradiction that D is not big. Since D is nef, $D^d = 0$.

Setup.

- Fix an ample Cartier divisor A on X.
- Fix $\epsilon \in \mathbb{Q}_{>0}$ sufficiently small. For $n \geq 1$, put

$$a_n = ((D + A/n)^d)^{1/d}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$

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Then P_n is ample and rational Cartier, $r_n \to \infty$, and $P_n^d \to \epsilon^d$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \le a_n/2$ for all sufficiently large n. The sequence index n is independent of the dimension d.

- Let $p_i : X \times X \to X$ be the projections. Put

$$Z = \mathbb{P}_{X\times X}\big(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)\big), \qquad \pi : Z \to X \times X.$$

Let S_1, S_2 be the two summand sections, and set

$$U = \pi^{-1}(X_{\mathrm{reg}} \times X_{\mathrm{reg}}) \setminus (S_1 \cup S_2).$$

Thus U is smooth and $\dim Z = 2d + 1$.

- Set $\xi = c_1(\mathcal{O}_Z(1))$. For an integer $s > 0$, put

$$R_{n,s} = (p_1 \circ \pi)^* P_n + (p_2 \circ \pi)^* P_n + s\xi.$$

We use the convention $\pi_*\mathcal{O}_Z(a) = \mathrm{Sym}^a(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D))$ for $a \ge 0$. Whenever $|kR_{n,s}|$ is used, $k > 0$ is an integer with kP_n Cartier.

- Choose ϵ sufficiently small that

$$(14\epsilon)^d < \tfrac{1}{4}, \qquad 80\epsilon < \tfrac{3}{4}.$$

The integer s is chosen after ϵ and before n tends to infinity. The multiple k is chosen after n.

Jet separation. Let V be a variety, let B be a Cartier divisor, and let $z \in V_{\mathrm{reg}}$. For $j \ge 0$, $|B|$ separates j-jets at z if

$$\forall u \in \mathcal{O}_{V,z}(B) \quad \exists \sigma \in H^0(V, \mathcal{O}_V(B)) \quad \sigma_z - u \in \mathfrak{m}_z^{j+1}\mathcal{O}_{V,z}(B).$$

Under the above assumptions, we use the following two lemmas.

Lemma 1.1. *Assume* (I_d). *Let X, D and $R_{n,s}$ be as in Setup. There exists an integer $s > 0$ such that, for every $n \gg 0$, $|kR_{n,s}|$ separates $(2d+2)k$-jets at a very general point $z \in U$, for all sufficiently divisible positive integers k.*

This is the large-Seshadri-constant construction of [1, Section 5], followed by the jet conversion in [1, Section 6.7].

Lemma 1.2. *Assume* (I_d). *Let X, D and $R_{n,s}$ be as in Setup. For every integer $s > 0$ and every $n \gg 0$, $|kR_{n,s}|$ does not separate $(2d+2)k$-jets at a very general point $z \in U$, for any positive integer k with kP_n Cartier.*

The terminal application in [1, Section 6.7] establishes the obstruction on a smooth resolution $w : \overline{X} \to X$. There $K_{\overline{X}} \sim E$ for an effective w-exceptional divisor E. An ample perturbation of w^*P_n supplies the complete-intersection flag, and the jets transfer on the isomorphism locus. For one fixed collection of complex data, the Frobenius determinant satisfies

$$\frac{3}{4}p^{2d} \leq \operatorname{ord}(\sigma) \leq \big(80\epsilon + O(p^{-1})\big)p^{2d},$$

which is impossible for arbitrarily large residual characteristics p.

Fix s supplied by Lemma 1.1, and choose n sufficiently large for both lemmas. For a sufficiently divisible k, Lemma 1.1 gives separation at a very general point of U, whereas Lemma 1.2 forbids it. This contradiction proves that D is big. □

Remark. The condition $K_X \sim 0$ is imposed at the outset. No adjoint reduction, index-one cover or terminalization is needed before Setup, and no irregularity assumption is imposed. Smooth projective varieties with $K_X \sim 0$ are included. *From nonvanishing to bigness.* If nonvanishing is already known, bigness also follows from the log-abundance input in (l_d). Choose an effective $\mathbb{Q}$-divisor $G \sim_{\mathbb{Q}} D$ and a sufficiently small rational $\tau > 0$ such that $(X, \tau G)$ is klt. Then

$$K_X + \tau G \sim_{\mathbb{Q}} \tau D \quad \text{is nef.}$$

Log abundance makes D semiample. A morphism defined by a free multiple of D has zero-dimensional general fibre: otherwise a curve in a fibre through a very general point would have D-degree zero, contrary to $\mathrm{n}(X, D) = d$. Thus the morphism is generically finite onto its image, and D is big.

Draft status. This is a conditional framework following [1, Sections 5–6]. The source inputs (l_d) and the inherited geometric and Frobenius arguments have not been independently verified here. The two jet lemmas record the source proof obligations; their complete proofs are not supplied by this framework.

References

[1] *Numerical Semiampleness of Nef Adjoint Divisors*, OpenAI mathematics collection, October 3, 2026, Sections 3–6. Source manuscript.